DESIGN TECHNIQUES

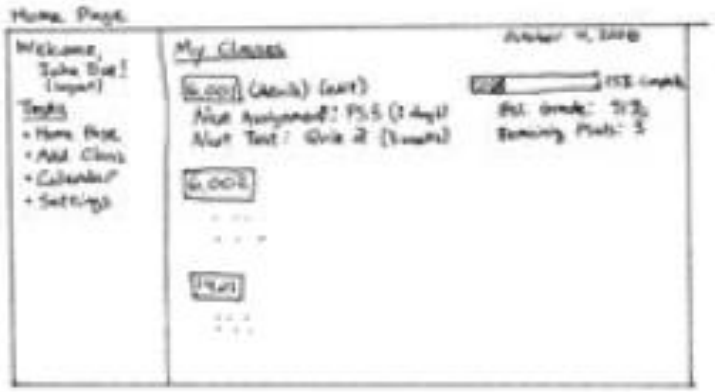
Design Techniques

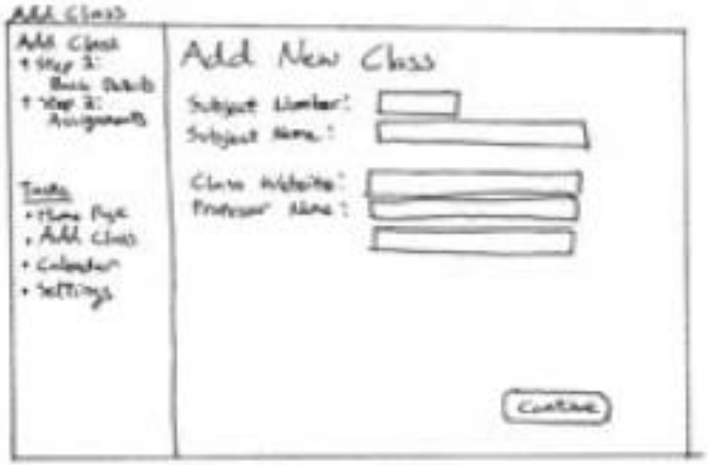
- Sketching
- Scenarios
- Storyboards

Sketching Designs

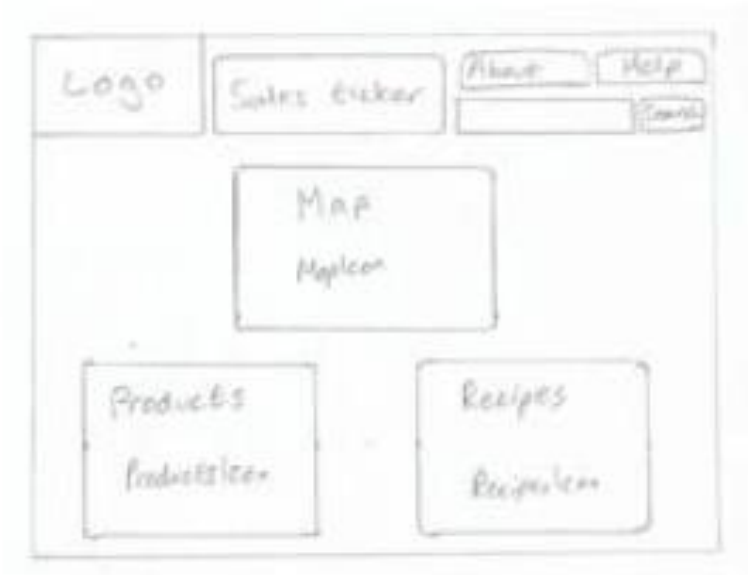
- Draw your ideas
  - By hand, not by drawing program
  - Paper or whiteboard
- Generate many ideas
- Design in a group

Examples of Design Sketches

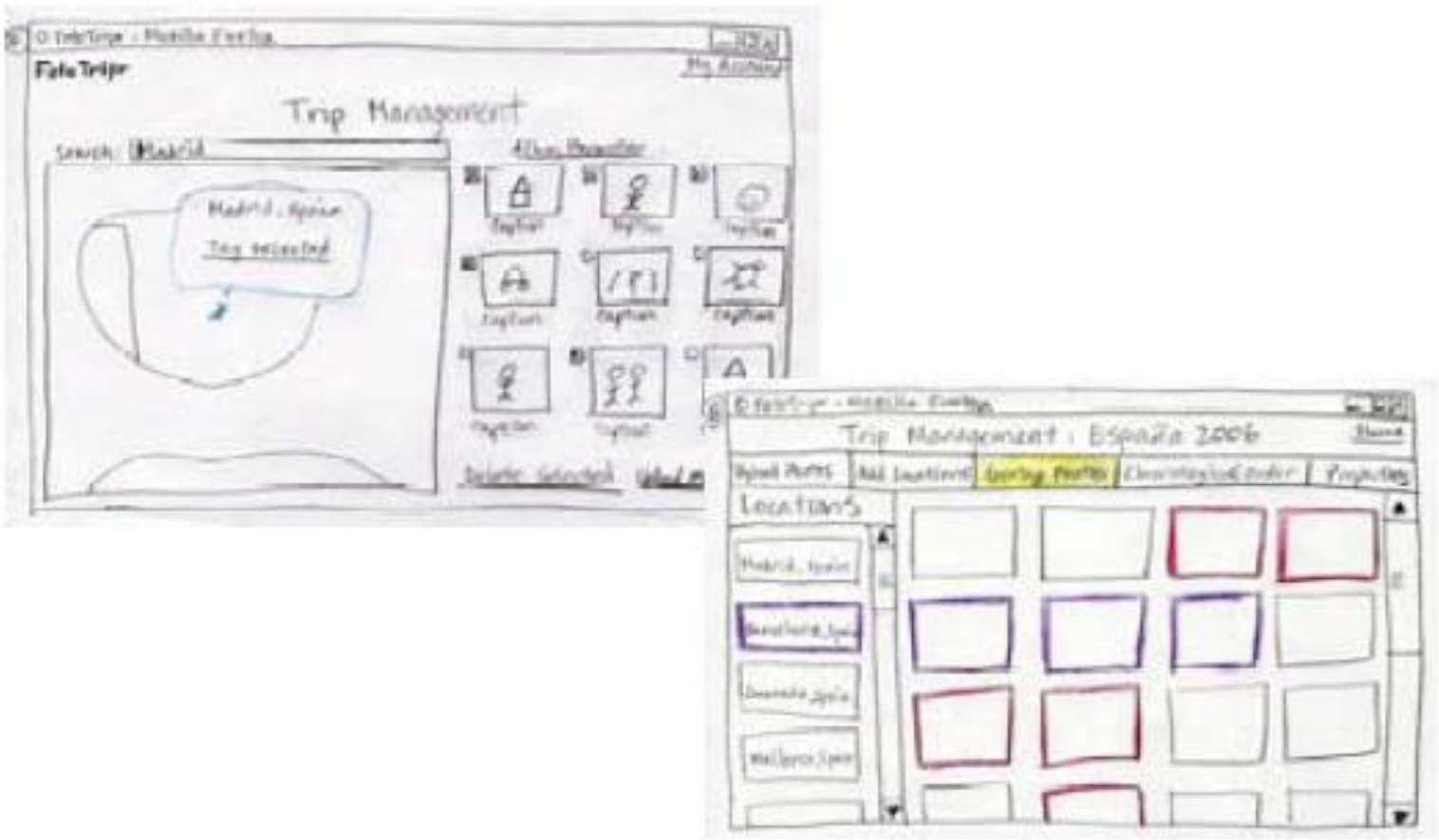




Examples of Design Sketches



Examples of Design Sketches



Scenarios

- Scenarios is a story about a user using the system
  - Concrete and realistic
  - Involves a user with a goal
  - Follows how the user achieves the goal

Scenarios …

- What will users want to do?
- Step-by-step walkthrough
  - What can they see (sketches, screen shots)
  - What do they do (keyboard, mouse etc.)
  - What are they thinking?
- Use and reuse throughout design

Scenario

Scenario - Example

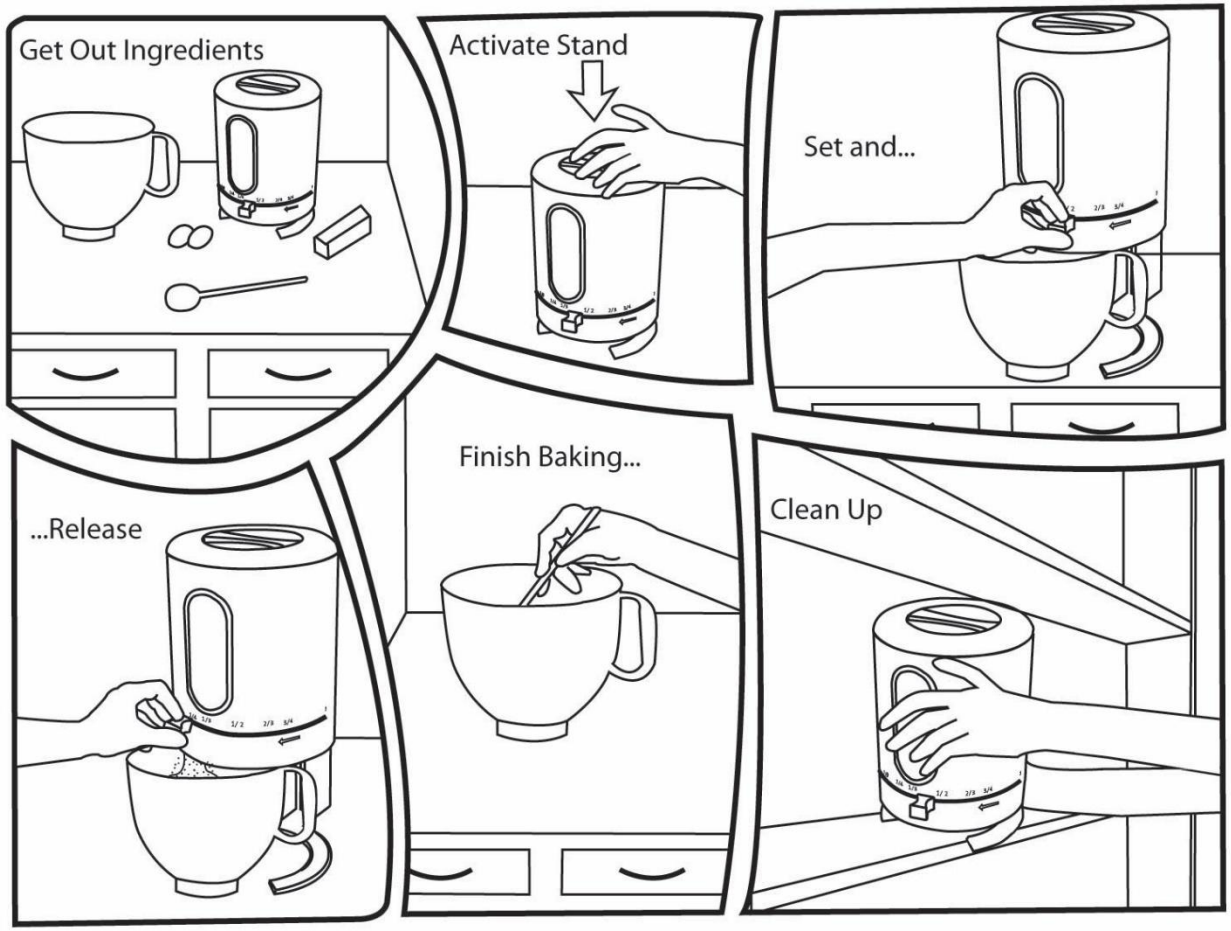
- A scenario is a little story
- It describes one possible course of events:

- Customer puts ATM card into machine and types password
- System validates the card and password
- System prompts the customer for a transaction type and amount
- Customer selects "withdraw $100 from checking"
- System checks with the central bank make sure customer has sufficient funds
- System dispenses the cash, returns card, and prints receipt
- System asks customer if he/she would like another transaction

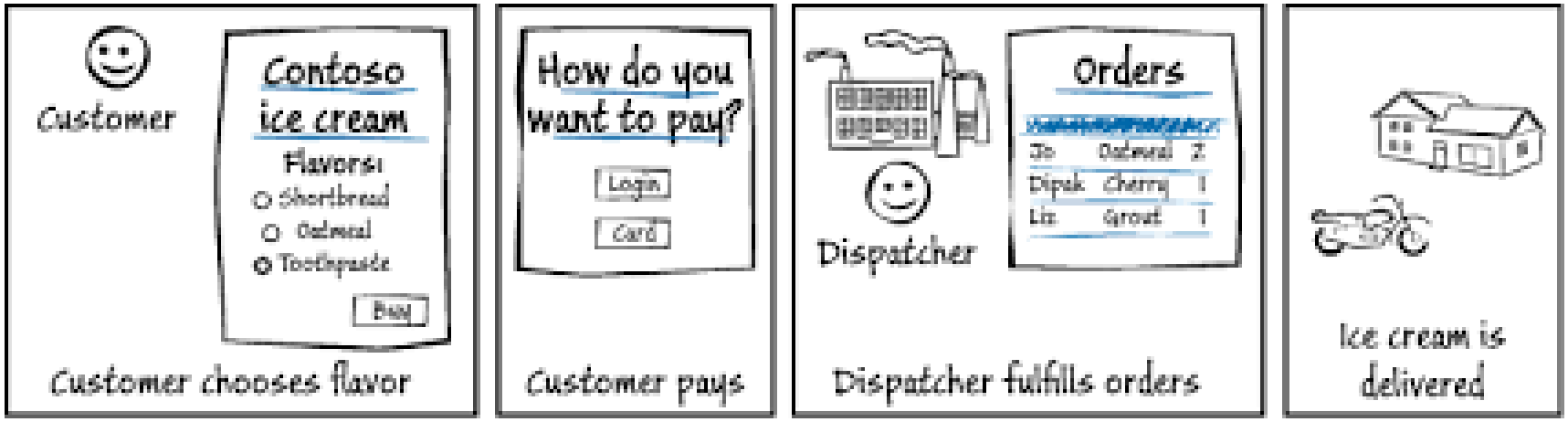
Storyboards

- Sequence of sketches illustrating a scenario
- First prototype of a design
  - Shows how the design can actually be used to satisfy a goal

Example-Story Board



Example-Story Board



SCREEN DESIGN AND LAYOUT

Basic principles

- Ask
  - What is the user doing?
- Think
  - What information, comparisons, order
- Design
  - Form follows function

Available tools

- Grouping of items
- Order of items
- Decoration - fonts, boxes etc.
- Alignment of items
- white space between items

Grouping and structure

logically together $\Rightarrow$ physically together

Billing details:
Name
Address: …
Credit card no

Delivery details:
Name
Address: …
Delivery time

Order details:

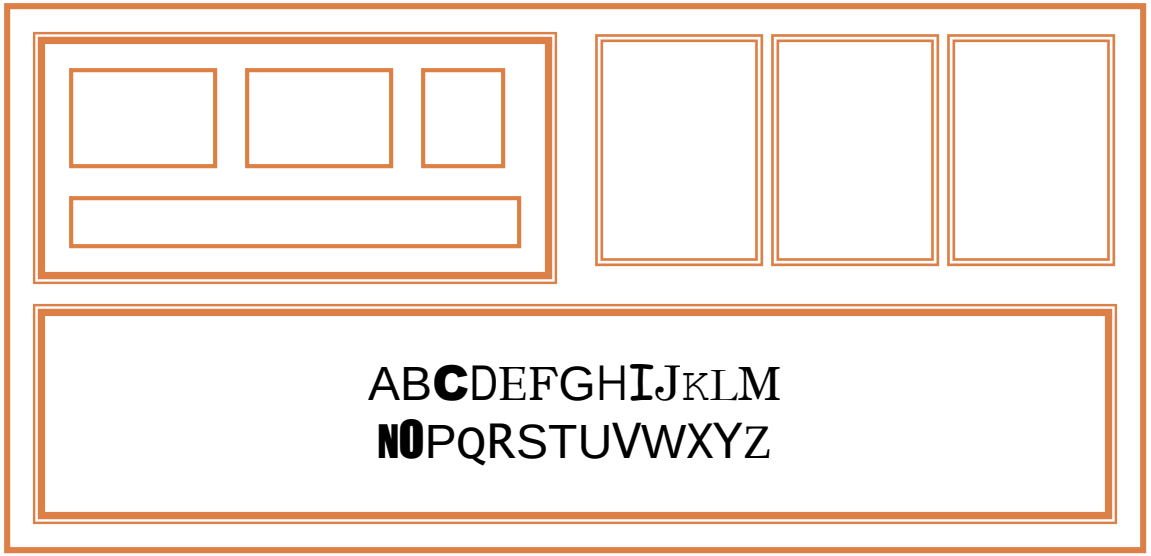
item	quantity	cost/item	cost
size 10 screws (boxes)	7	3.71	25.97
……	…	…	…

Order of groups and items

- think! - what is natural order
- should match screen order!
  - use boxes, space etc.
  - set up tabbing right!
- instructions
  - beware the cake recipie syndrome!
    … mix milk and flour, add the fruit
    after beating them

Decoration

- use boxes to group logical items
- use fonts for emphasis, headings
- but not too many!!



Alignment - text

- you read from left to right (English and European)

  $\Rightarrow$ align left hand side

Willy Wonka and the Chocolate Factory
Winston Churchill - A Biography
Wizard of Oz
Xena - Warrior Princess

boring but readable!

fine for special effects but hard to scan

Willy Wonka and the Chocolate Factory
Winston Churchill - A Biography
Wizard of Oz
Xena - Warrior Princess

Alignment - names

- Usually scanning for surnames make it easy! ⇒

Umarani Jayaraman
Masilamani V
Noor Mahammad
Sadagopan N
Sivaselvan B

Jayaraman , Umarani
V, Masilamani
Mahammad, Noor
N, Sadagopan
B, Sivaselvan

Umarani	Jayaraman
Masilamani	V
Noor	Mahammad
Sadagopan	N
Sivaselvan	B

Alignment - numbers

think purpose!

which is biggest?

532.56
179.3
256.317
15
73.948
1035
3.142
497.6256

Alignment - numbers

visually:

long number = big number

align decimal points

or right align integers

```
  627.865
    1.005763
  382.583
 2502.56
  432.935
    2.0175
  652.87
   56.34
```

Multiple columns

- scanning across gaps hard:

(often hard to avoid with large data base fields)

sherbert	75
toffee	120
chocolate	35
fruit gums	27
coconut dreams	85

multiple columns - 2

- use leaders

sherbert	75
toffee	120
chocolate	35
fruit gums	27
coconut dreams	85

Multiple columns - 3

- or greying (vertical too)

sherbert	75
toffee	120
chocolate	35
fruit gums	27
coconut dreams	85

Multiple columns - 4

- or even (with care!) ‘bad’ alignment

sherbert	75
toffee	120
chocolate	35
fruit gums	27
coconut dreams	85

white space - the counter

WHAT YOU SEE

white space - the counter

WHAT YOU SEE

THE GAPS BETWEEN

space to separate

space to structure

space to highlight

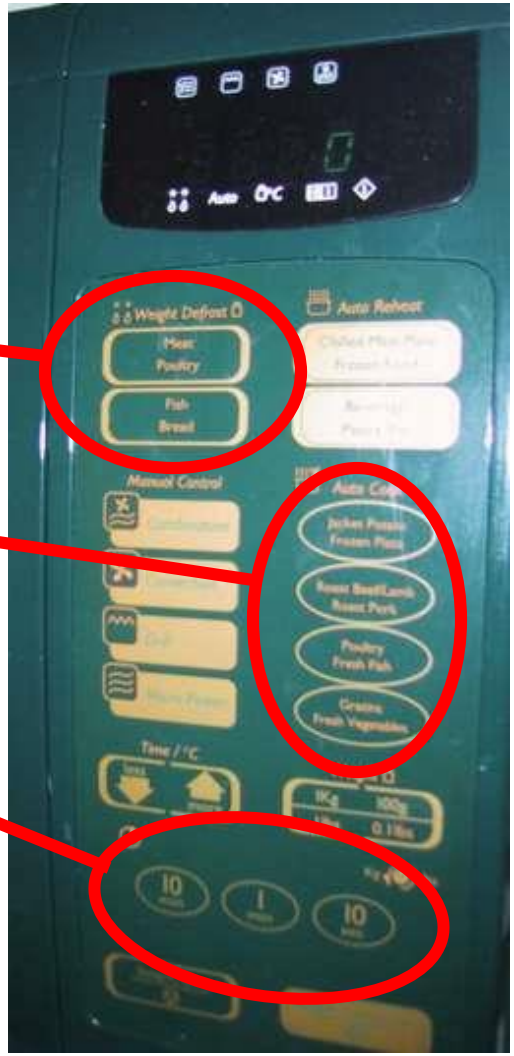
Physical controls

- grouping of items
  - defrost settings

defrost settings

type of food

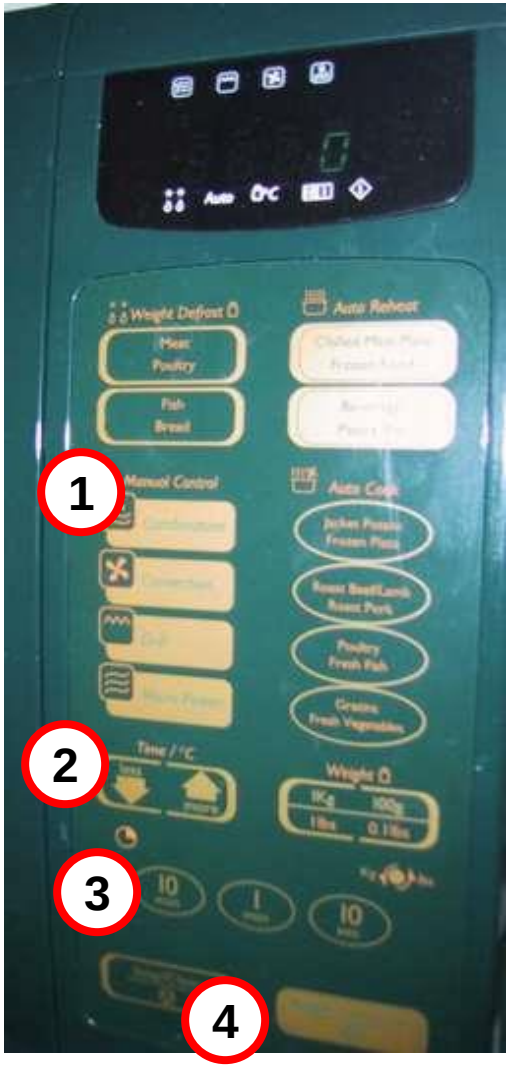
time to cook



Physical controls

- grouping of items
- order of items
  1) type of heating

1) type of heating
2) temperature
3) time to cook
4) start

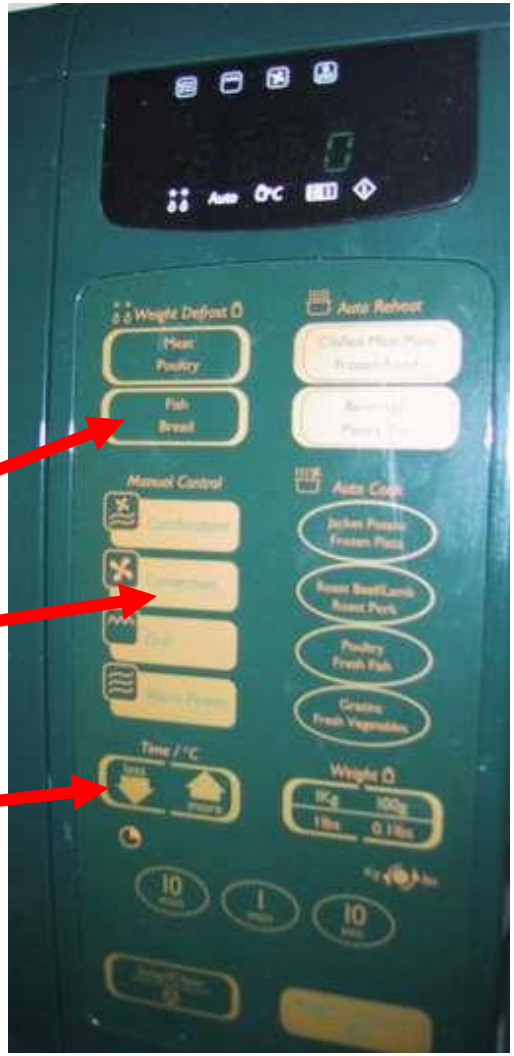


Physical controls

- grouping of items
- order of items
- decoration
  - different colours

    different colours for different functions

    lines around related buttons (temp up/down)

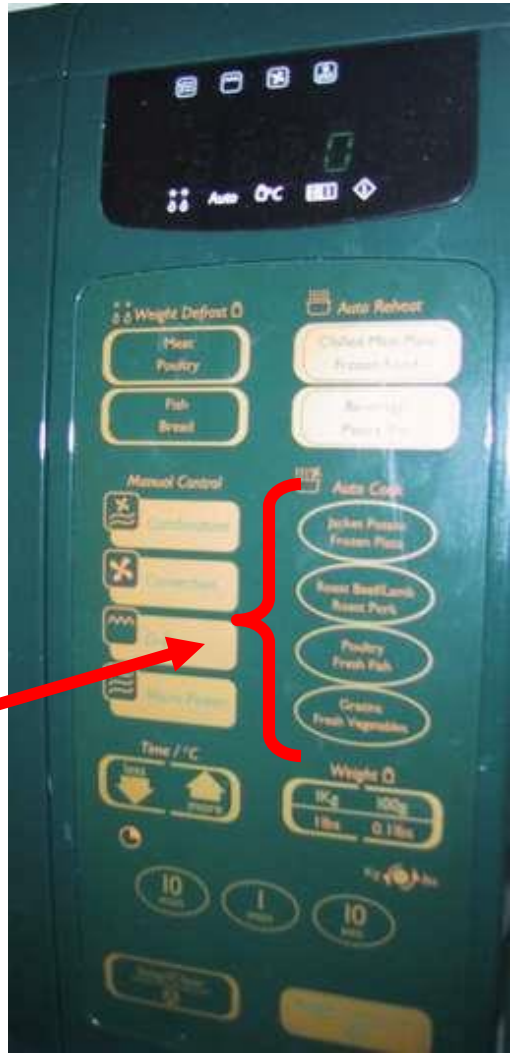


Physical controls

- grouping of items
- order of items
- decoration
- alignment
  - centered text in buttons

centred text in buttons

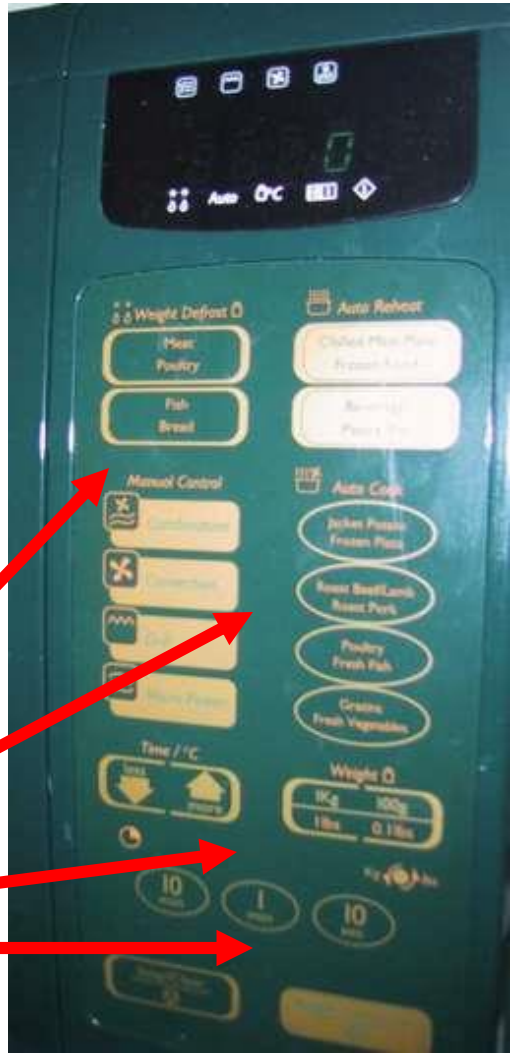
? easy to scan ?



Physical controls

- grouping of items
- order of items
- decoration
- alignment
- white space
  - gaps to aid grouping

gaps to aid grouping



DESIGN PATTERNS IN HCI

So far

- What is design?
- Design process
- Design techniques
- Design principles
- Today
  - Design patterns

Design patterns

- Patterns are good solutions to common problems
- What is pattern?
  - **Decorative Design**
- There are several kinds of patterns
  - Structural
  - Navigation
  - Widgets

Structural

- Wizard
  - sequence of dialog boxes with Next and Previous buttons
- Center stage
  - word processor, drawing programs
- High-density information display
  - mail and google maps

Navigation

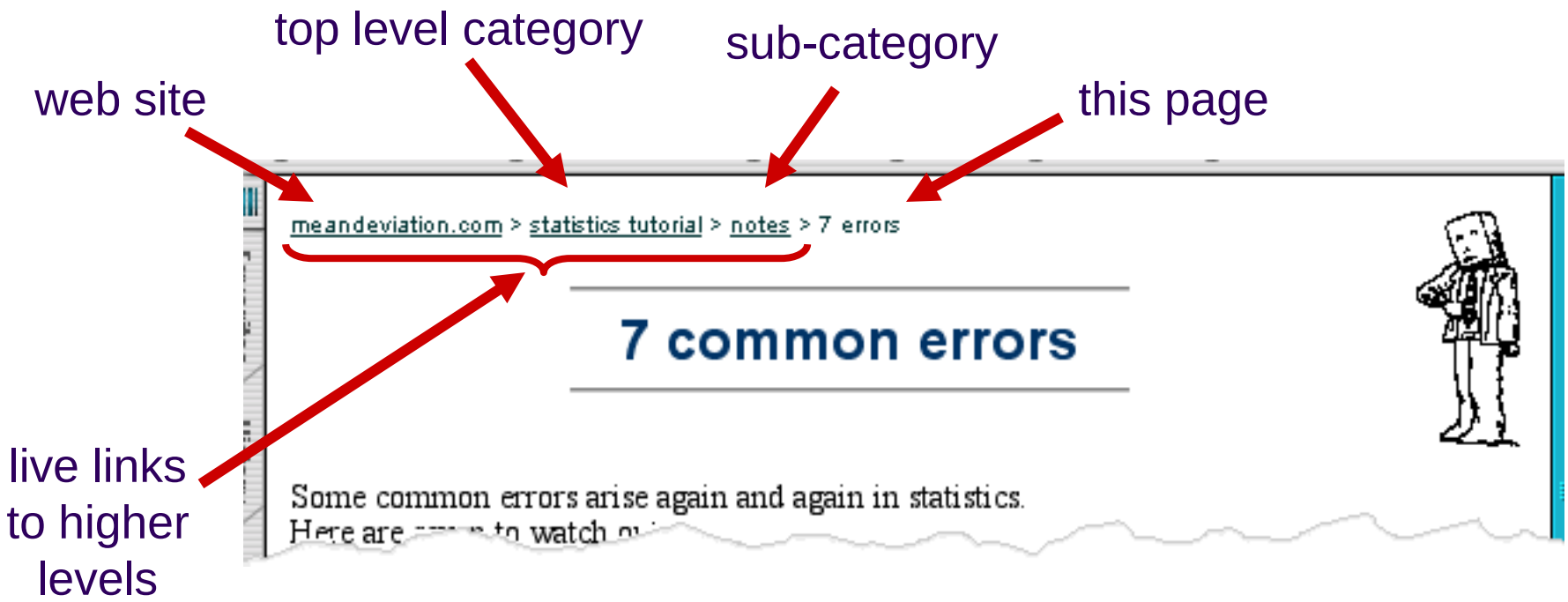
- Pagination
- Breadcrumbs (where you are)

Pagination

- **Pagination** divides up a long list of data into pages, with a control showing the page numbers (e.g. 1 2 3 4 ... 103) as hyperlinks.

Breadcrumbs (where you are)

shows path through web site hierarchy



Widgets for 1-of-N Choices

Radio buttons

Drop-down menu

Single-selection listbox

Toggle buttons

Radio Buttons **for 1-of-N Choices**

- Radio buttons are the conventional idiom for making 1-of-N selections.
- They are fast
- All the choices are visible
- It uses a lot of screen real estate as N gets large

Toggle Buttons for 1-of-N Choices

- Toggle buttons can offer larger targets than radio buttons (faster hit)

- Toggle buttons are often used for mode switches, or for 1-of-N choices in toolbars.

Drop-down menu for 1-of-N Choices

- A **single-selection listbox** is a static version of the same thing, where the list of N choices is always visible, and the chosen item is indicated by a highlight.
- Drop-down menus are very compact, but require two pointing operations (point and click to open the list, then point to the new choice)

- All these widgets are useful for 1-of-N choices,

In design Issues:

- Avoid using N checkboxes for 1-of-N choices.
- Radio buttons are a visually distinct widget that communicates the affordance that the choices are exclusive (i.e., that you can only select one at a time).
- Checkboxes fail to convey this.

Widgets for 1-of-2 Choices

- For the special case where N=2, you can also use a single **checkbox** or **toggle button.**

Widgets for 1-of-2 Choices

- Widgets for 1-of-N choices (with N=2), plus:

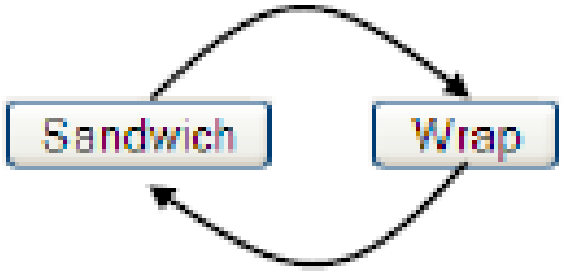
Checkbox ☑ Toasted

Toggle button [Toasted]

- Avoid:

☐ Sort in ascending order

☐ Insert vs. overstrike



Widgets for K-of-N Choices

N checkboxes

Multiple-selection listbox

Two listboxes

Widgets for K-of-N Choices

- **Checkboxes** are a natural way to do this.
- **A multiple-selection listbox** is another way to do it.
  - Lack of learnability- multiple selections are not visible to users
  - Errors(the selection is very fragile) - one accidental click can clear a carefully-created selection
- **Two listboxes** are still another way
  - 'K' choices you have made clearly visible
  - Less fragile
  - It uses wider screen

Widgets for K-of-N Choices

- **Here's a design question: when the user moves an item over to the 'K' list, should it disappear from the N list?**

Widgets for K-of-N Choices

- **Here's a design question: when the user moves an item over to the 'K' list, should it disappear from the N list?**
- Maybe not disappear, since that would remove visual search landmarks from the N list

Widgets for K-of-N Choices

- **Here's a design question: when the user moves an item over to the 'K' list, should it disappear from the N list?**
- Maybe not disappear, since that would remove visual search landmarks from the N list
- But some indication in the N list of which items have already been selected would improve the visibility of system status and reduce errors (like selecting the same item again).

levels

- widget choice
  - menus, buttons etc.
- screen design
- navigation design
- environment
  - other apps, O/S

Website…

- widget choice
- screen design
- navigation design
- environment

- elements and tags and links- `<a href="...">`
- page design
- site structure
- the web, browser, external links

Physical devices

- widget choice
- screen design
- navigation design
- environment

- controls
  - buttons, knobs, dials
- physical layout
- Main modes of device
- the real world

Poor interface design

- Increased mistakes in
  - Data entry
  - System operation
- Inaccessible functionality
- User frustration
  - Low productivity and/or
  - Under-utilization
- System failure because of user rejection

GOLDEN RULES OF DESIGN

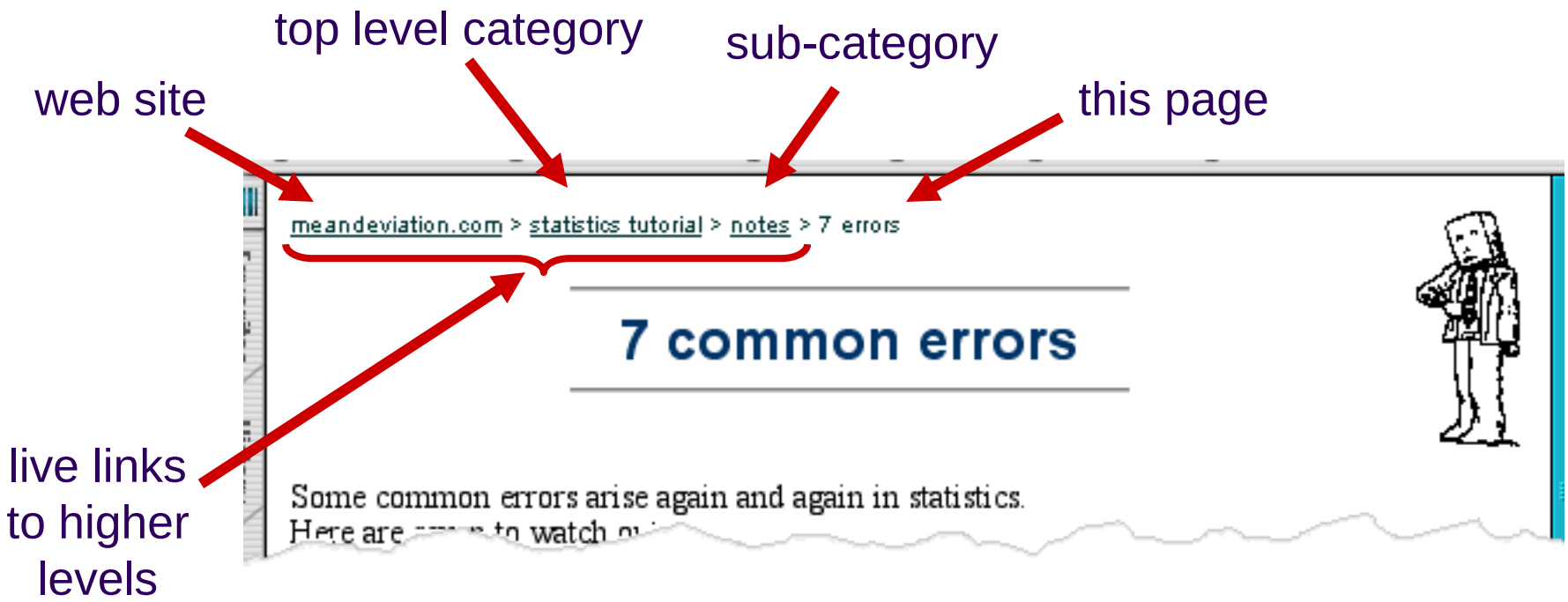
from one screen looking out

Four golden rules of design

- knowing where you are
- knowing what you can do
- knowing where you are going (with respect to goal)
  - or what will happen
- knowing where you've been (with respect to goal)
  - or what you've done

Knowing where you are – breadcrumbs

shows path through web site hierarchy

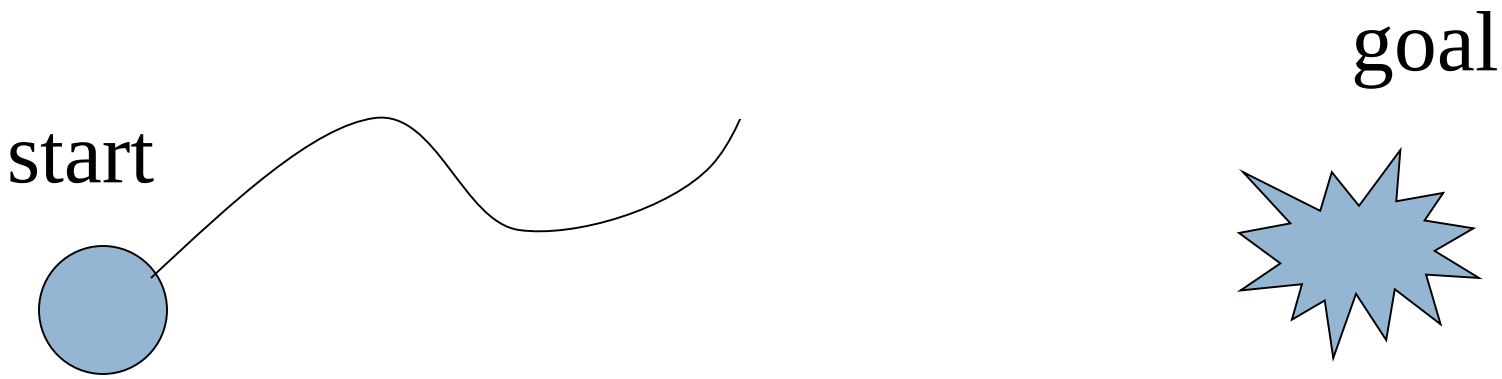


knowing what you can do

- What is active what is passive
  - where do you click
  - where do you type
- consistent style helps
  - e.g. web underlined links
- Labels and icons
  - standards for common actions
  - language – bold = current state or action

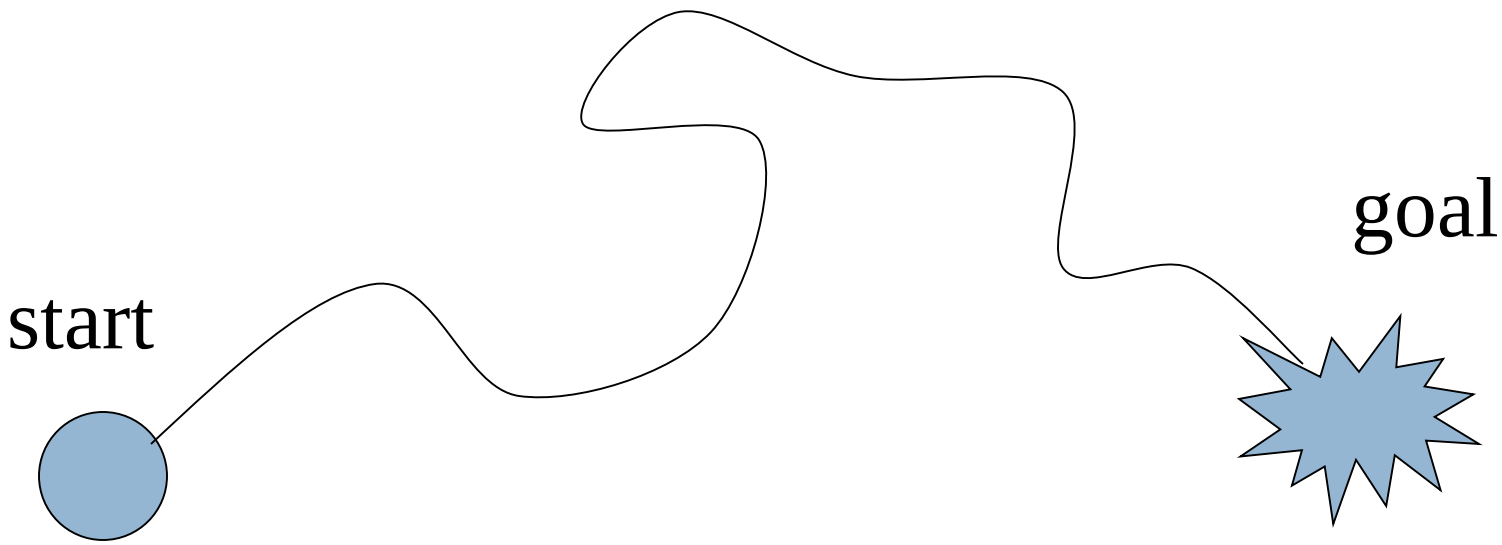
knowing where you are going goal seeking

knowing where you are going
goal seeking



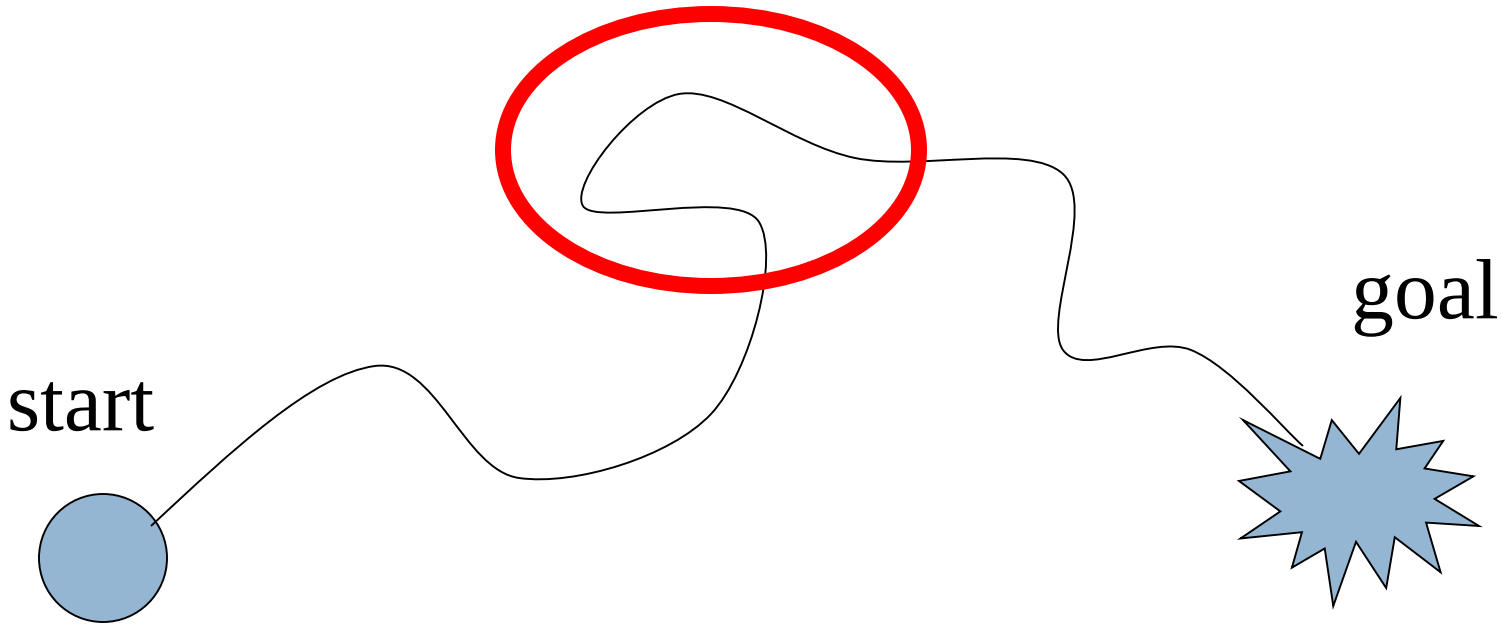
progress with local knowledge only ...

knowing where you've been- goal seeking



… but can get to the goal

knowing where you've been- goal seeking



… try to avoid these bits!

USER ACTION AND CONTROL

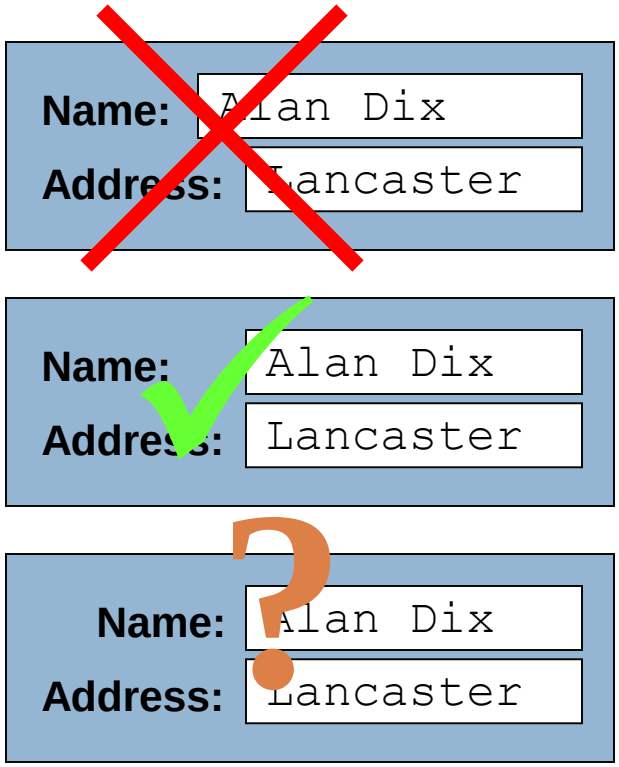
entering information

knowing what to do

affordances

Entering information

- Forms, dialogue boxes
  - Presentation + data input
  - Similar layout issues
  - Alignment - N.B. different label lengths
- Logical layout
  - Use task analysis
  - Groupings
  - Natural order for entering information
    - Top-bottom, left-right (depending on culture)
    - Set tab order for keyboard entry



Presenting information

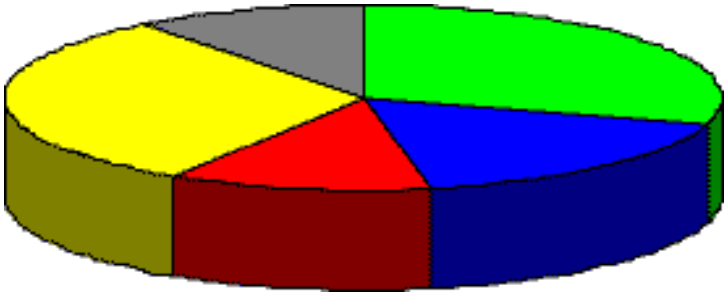
- Purpose matters
  - Sort order (which column, numeric alphabetic)
  - Text vs. diagram
  - Scatter graph vs. histogram
- Use paper presentation principles!
- But add interactivity
  - Softens design choices
    - e.g. re-ordering columns

name	size
chap10	12
chap5	16
chap1	17
chap14	22
chap20	27
chap8	32
...	...

Colour and 3D

- Both often used very badly!
- Colour
  - Older monitors limited palette
  - Colour over used because 'it is there'
  - Beware colour blind!
  - Use sparingly to reinforce other information
- 3D effects
  - Good for physical information and some graphs
  - But if over used …

    e.g. text in perspective!! 3D pie charts



bad use of colour

- over use - without very good reason (e.g. kids' site)

- over use - without very good reason (e.g. kids' site)

- colour blindness
- poor use of contrast
- do adjust your set!
  - adjust your monitor to greys only
  - can you still read your screen?

Across countries and cultures

- localisation & internationalisation
  - changing interfaces for particular cultures/languages
- globalisation
  - try to choose symbols etc. that work everywhere

- simply change language?
  - use ‘resource’ database instead of literal text
    … but changes sizes, left-right order etc.
- deeper issues
  - cultural assumptions and values
  - meanings of symbols

    e.g tick and cross … +ve and -ve in some cultures
    … but … mean the same thing (mark this) in others